YARA-BASED THREAT DETECTION

Safe Sample Detection Lab

PROJECT DOCUMENTATION & TEST EVIDENCE

Environment: Windows Command Prompt • YARA 4.4.0
Sample type: Harmless plain-text files with artificial indicators
Scope: Rule-logic validation only; no real malware used

1. Executive Summary

This project demonstrates a basic YARA detection workflow using harmless plain-text training files. Two rules were created: one requires a combination of artificial indicators, while the second checks for a single artificial string. The supplied terminal evidence shows both rules matching the suspicious-like training file, no match output for the clean control file, and the same matches during a recursive scan.

These results confirm the rules' expected behavior on the provided examples only. They do not demonstrate detection of real malware.

2. Objectives

- Run YARA from Windows Command Prompt.
- Understand the relationship between a YARA executable, rule file, and target file.
- Create rules using string indicators and logical conditions.
- Compare a positive training sample with a clean control sample.
- Scan a directory recursively and record the results.
- Document the process and its limitations accurately.

3. Project Structure

- `safe_training_rule.yar` — two YARA training rules.
- `samples/sample_suspicious.txt` — plain text containing alpha, beta, and context markers.
- `samples/sample_clean.txt` — control text without the required marker combination.
- `samples/sample_one_indicator.txt` — follow-on test file containing only the alpha marker.
- `evidence/screenshots/` — selected screenshots arranged in the same order as the results section.
- `README.md` — short setup and reproduction guide.

4. Test Environment and Method

Tests were run from Windows Command Prompt using the YARA executable at `C:\Users\KIIT\Downloads\yara-v4.4.0-2250-win64\yara64.exe`. The project folder contained the rule file and a `samples` directory. Only plain-text training files were used.

YARA prints a rule name and target path when a rule matches. If no rule matches a target, the command returns without a detection line. The `-r` option enables recursive scanning of a directory.

5. Detection Rule Logic

5.1 Simulated_Threat_Indicator_Detection

```
Condition: 2 of ($alpha, $beta) and $context
```

The rule requires both strings in the two-item group (`$alpha` and `$beta`) because the condition asks for 2 of 2, and it also requires `$context`. All required elements must be present for this rule to match.

5.2 Single_Indicator_Test

```
String: $test = "simulated_indicator_alpha"
Condition: $test
```

This rule matches when the artificial alpha string is present. It is intentionally simpler than the first rule to demonstrate the difference between a single-indicator rule and a multi-indicator condition.

6. Test Commands

```
"C:\Users\KIIT\Downloads\yara-v4.4.0-2250-win64\yara64.exe" safe_training_rule.yar samples\sample_suspicious.txt
"C:\Users\KIIT\Downloads\yara-v4.4.0-2250-win64\yara64.exe" safe_training_rule.yar samples\sample_clean.txt
"C:\Users\KIIT\Downloads\yara-v4.4.0-2250-win64\yara64.exe" -r safe_training_rule.yar samples
```

7. Results

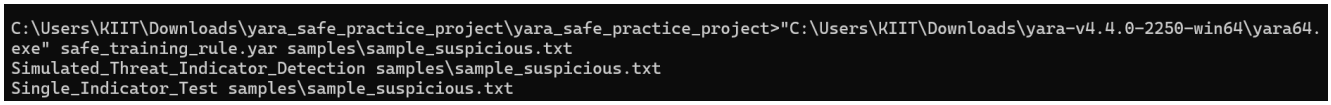
Test	Expected	Observed	Assessment
Individual scan: suspicious-like sample	Both rules match	Both rule names printed	Pass
Individual scan: clean control	No match	No rule name printed	Pass
Recursive scan: samples folder	Matching files reported	Both rules printed for sample_suspicious.txt	Pass
One-indicator follow-on test	Single_Indicator_Test only	Not shown in submitted screenshot evidence	Not verified

The observed positive and negative examples are consistent with the intended rule logic. The test set is small and deliberately constructed; it cannot establish a production false-positive rate.

8. Screenshot Evidence

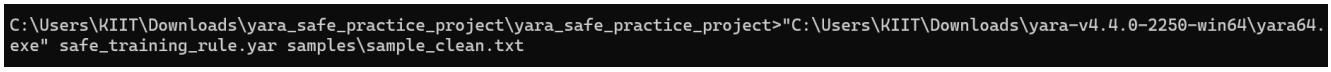
The evidence below is arranged in test sequence: individual positive match, clean control, then recursive scan. Only one representative screenshot is included for each distinct result.

Figure 1 — Individual scan of the suspicious-like training sample



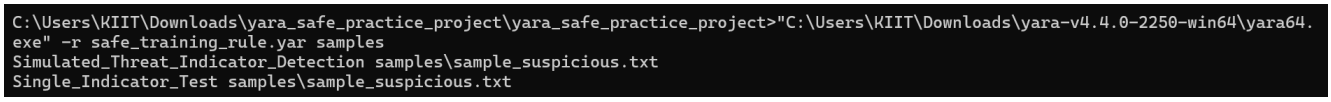
Both Simulated_Threat_Indicator_Detection and Single_Indicator_Test matched sample_suspicious.txt.

Figure 2 — Clean control sample



The command targeted sample_clean.txt and produced no rule-match line.

Figure 3 — Recursive scan of the samples directory



The -r scan reported both matching rules for sample_suspicious.txt.

9. Limitations and Responsible Interpretation

- All indicators are artificial strings in plain-text files; no real malware was used.
- The project validates rule behavior on constructed examples, not real-world malware detection.

- The small sample set is not representative enough to measure a meaningful false-positive rate.
- The one-indicator follow-on test is included as a practice file, but its result was not present in the supplied screenshots.

10. Conclusion

The lab demonstrates the core YARA workflow: define strings, express detection logic, run rules against files, compare positive and negative cases, and record terminal evidence. No real malware analysis performed.